

Victor Gladysz

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ABOUT

Dedicated Mechanical Engineering student at the University of Southampton, with hands-on experience in manufacturing, vehicle dynamics, powertrain simulation, and mechanical design. As an active member of the University's Formula Student team, I am currently serving as Head of Powertrains, building simulations and tools to improve the design intent of our electric powertrain.

EXPERIENCE

Head of Powertrains

Jul 2025 – Present

University of Southampton Formula Student Team

Southampton, UK

- Simulated powertrain performance by creating a data-fitting method in MATLAB to tune model parameters, improving the alignment between predicted and measured data.
- Developed a motor model to enhance an existing point-mass lap time simulation tool, improving prediction accuracy against real-world performance.
- Wrote a script to minimise driveshaft plunge, optimising the location of the differential, reducing the wear on tripod CV joints.
- Utilized Ansys FEA and Excel to allow for more design iterations in the same amount of time, streamlining the design process.
- Implemented both on-track data and experimental results, into the design of a new cooling loop, saving weight and more efficiently cooling the electric motor and inverter.

Suspension Team Member

Sep 2024 - Jul 2025

University of Southampton Formula Student Team

Southampton, UK

- Employed the Lavenberg-Marquardt algorithm in Matlab, to simulate suspension travel, allowing for clash checking between components and vehicle dynamics analysis.
- Presented during the Cost and Manufacturing event at FSUK 2025, demonstrating an advanced understanding of manufacturing methods, and contributing to the team placing 13th in the event, and 10th overall.
- Developed a SolidWorks macro using Granta Edupack and VBA to automatically set custom part properties, including cradle-to-gate carbon emissions and expected costs.
- Gained hands-on experience with ANSYS FEA, performing simulations on suspension components to test their performance under various loading conditions.

Lifeguard

Jun 2022 – Sep 2025

Palmyra Community Pool

Palmyra, PA, USA

- Directed fellow lifeguards to deliver high quality swim lessons, enhancing communication between lifeguards and students' parents.
- Maintained and monitored the water filtration systems to ensure uninterrupted operations, responding on short notice when required.

Mechanical & Electrical Lead

Aug 2020 – Jul 2024

FIRST Robotics Competition Team 2539: The Krypton Cougars

Palmyra, PA, USA

- For four consecutive years, I directed the creative design and development of unique and competitive robots under strict budget and time constraints.
- Provided crucial team leadership, overseeing the maintenance of an alliance of four robots at the World Championship in 2022, 2023, & 2024.

EDUCATION

University of Southampton

Southampton, UK

Master of Engineering in Mechanical Engineering, Expected Graduation: July 2029

Sep 2024 – Present

- Coursework: Statics, Dynamics Materials, Thermodynamics, & Electronics.
- Societies: Formula Student, Engineering Society, Robotics Outreach, & Swimming Club.

Palmyra Area High School

Palmyra, PA, USA

High School Diploma, Highest Honours

Aug 2020 – Jun 2024

- Societies: FIRST Robotics, Varsity Swimming Captain, Scouting America: Eagle Scout & National Honour Society.

PROJECTS

Cost & Carbon Estimation | *Granta Edupack, Go, SQL, Excel*

Nov 2024 – Jul 2025

- Developed a REST API using Go, that allowed database indexing used for the automated generation of a Costed Carbonized Bill of Materials.
- Implemented a MySQL database, that contained part information, used to calculate environmental impacts of the material, transportation, and manufacturing methods.

FEEG1201 Group Project | *Solidworks, Python*

Mar 2024 – Apr 2025

- Led a team of first year engineers to design a STEM exhibit to teach students about energy conservation.
- Designed a model roller-coaster, demonstrating the conversion of gravitation potential to kinetic energy. The design allowed for simple construction with limited tools and resources.
- Simulated the final velocity of the coaster using Python, taking into account rolling resistance.
- Presented the final design to a group of lecturers, earning the design a place in the university's design show.

RoboHack 2025 | *Python*

Feb 2025

- Creatively interpreted the provided rules, to extract an unwritten advantage, simplifying the requirements, and winning the tournament.
- Developed a localization algorithm, challenging myself, and allowing the robot to better evaluate its environment.
- Rapidly adapted the match strategy when faced with electrical failures, and after evaluating competitors' performance.
- Worked with students from different disciplines to design a robot to compete in a round robin tournament.

ACTIVITIES & HOBBIES

Swimming

Aug 2018 – Present

- Secured sponsorships through professional relationships with local companies to fund raise for upgraded high school swim team equipment.
- As a co-captain, led a team of swimmers to win the Keystone AAA Division.
- Joined the university swim club to remain in shape and make new friends, while still balancing my course load and other commitments.

Scouting America

Apr 2017 – May 2024

- Staffed National Youth Leadership Training, NYLT, for three years, teaching fellow scouts leadership skills, including effective communication and conflict resolution.
- Earned Eagle Scout, the highest honour in Scouting America, by leading a team of adult leaders, volunteers, and scouts to renovate a shed used by a local swim team.
- Served as Chapter Vice Chief in the Order of the Arrow: Scouting's National Honor Society, contributing to event planning and supporting leadership initiatives at the chapter, lodge, and section levels.
- Active member of Friends of Camp Mack, contributing to service projects and fundraising efforts to support long-term camp ground improvements.

SKILLS & TECHNOLOGIES

Skills: Strong Work Ethic, Team Player, Responsive Under Pressure, Problem Solver, Effective Communicator, Adaptable, Critical Thinker, Leader, Growth Mindset, Fluent Public Speaker, & Strong Time Management Skills

Technical Skills: Mathematical Modelling, Simulation Development, CAD, Engineering Drawings, Data Analysis & Visualisation, GD&T, & FEA

Technologies: SolidWorks, ANSYS (Workbench, Mechanical), Motec i2 Standard, & Microsoft Office Products (Excel, Word, PowerPoint, Power BI, & Power Automate)

Programming Languages: Matlab, Python, Go, LaTeX, VBA, & SQL

REFERENCES

Availabe upon request.